

Lecture Notes on Differential Equations and Transform Techniques

Lecture Notes for Spring 2025

The purpose of these lecture notes is to provide a concise summary of the key concepts covered in the course. They are designed to serve as a **quick reference** and should be used in conjunction with the core textbooks listed below. While some sections are adapted from these referenced texts, the notes are not intended to replace the original works, which offer more comprehensive explanations. To gain a deeper understanding, students are encouraged to refer to the textbooks listed.

References:

1. W. E. Boyce, R. C. DiPrima, D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 9th edition, Wiley, 1997.
2. Brad G. Osgood, *Lectures on Fourier Transform and Its Applications*, IAS/Park City Mathematics Series / American Mathematical Society (AMS), 2019.
3. Charles Henry Edwards, David E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th edition, Pearson, 2000.
4. George F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd edition, CRC Press, 2016.
5. Ruel V. Churchill, *Fourier Series and Boundary Value Problems*, McGraw-Hill, 1941.
6. Gustav Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Springer, 2012.
7. Morris Tenenbaum, Harry Pollard, *Ordinary Differential Equations*, Dover Books on Mathematics, Dover Publications, 1985.

EE1060: Differential Equations and Transform Techniques**Lecture Notes - 06****Topics covered :**

1. Substitution Method
2. Bernoulli Equation

- Consider a non-exact first-order ODE

$$P(x, y)dx + Q(x, y)dy \quad (1)$$

such that $P(x, y)$ and $Q(x, y)$ are homogeneous functions of the same degree. Specifically, they satisfy the conditions

$$P(\alpha x, \alpha y) = \alpha^N P(x, y) \text{ and } Q(\alpha x, \alpha y) = \alpha^N Q(x, y) \quad (2)$$

where N is the degree of homogeneity. Differential equations that satisfy this condition are often referred to as [Homogeneous first-order ODE](#).

- A non-exact homogeneous first-order ODE can be transformed into a separable differential equation using the [substitution](#) $u = \frac{y}{x}$. This substitution takes advantage of the homogeneity of the equation. Substituting $y = ux$ in (1) gives

$$P(x, ux)dx + Q(x, ux)d(ux) = 0 \implies P(x, ux)dx + Q(x, ux)(udx + xdu) = 0 \quad (3)$$

Since $P(x, y)$ and $Q(x, y)$ are homogeneous functions of degree N , (3) can be written as

$$x^N [P(1, u)dx + Q(1, u)(udx + xdu)] = 0 \quad (4)$$

For the equation to hold for $x \neq 0$, we have

$$P(1, u)dx + Q(1, u)(udx + xdu) = 0 \implies [P(1, u) + uQ(1, u)]dx + xQ(1, u)du = 0 \quad (5)$$

Equation (5) can be simplified as

$$\frac{Q(1, u)}{(P(1, u) + uQ(1, u))} \frac{du}{dx} = -\frac{1}{x} \quad (6)$$

which is a separable differential equation. The function $u(x)$ can be solved by integrating both sides of (6).

It is important to note that the solution to the original non-exact ODE is given by $y(x) = u(x)x$, where $u(x)$ is the solution to the separable equation (6).

- A first-order differential equation of the form

$$\frac{dy}{dx} + p(x)y = q(x)y^n \quad (7)$$

where n is a constant, and $p(x)$ and $q(x)$ are continuous functions of x is called the [Bernoulli's equation](#), which is non-linear for $n > 1$. A Bernoulli's equation [1] with $n > 1$ can be converted into a first-order linear ODE using the substitution

$$u = y^{1-n} \quad (8)$$

For the chosen substitution,

$$\frac{du}{dx} = (1-n)y^{-n} \frac{dy}{dx} \implies \frac{dy}{dx} = \frac{y^n}{(1-n)} \frac{du}{dx} \quad (9)$$

Using (8) and (9), (7) can be simplified as

$$\frac{y^n}{(1-n)} \frac{du}{dx} + p(x)y = q(x)y^n \implies \frac{du}{dx} + (1-n)p(x)y^{1-n} = (1-n)q(x) \implies \frac{du}{dx} + (1-n)p(x)u(x) = (1-n)q(x) \quad (10)$$

Equation 10 is a first-order linear ODE which can be solved using the integrating factor approach.

- **Example:** Solve the first-order ODE

$$\frac{dy}{dx} + xy = y^2 \quad (11)$$

The given ODE is a Bernoulli's equation with $n = 2$, which can be solved using the substitution $u = \frac{1}{y}$. Using the substitution $u = \frac{1}{y}$, transforms (11) to

$$\frac{du}{dx} - xu = -1 \quad (12)$$

Further the integrating factor $h = e^{\int -x dx} = e^{-\frac{x^2}{2}}$ transforms (12) to

$$\frac{d}{dx}(hu) = -h, \quad (13)$$

which is a separable DE. Integrating (13) on both sides gives

$$e^{-\frac{x^2}{2}} u = - \int e^{-\frac{x^2}{2}} dx \quad (14)$$

The integral on the right-hand side is a standard Gaussian integral, and we denote the result as $\Phi(x)$ (the error function). So,

$$u = e^{\frac{x^2}{2}} (-\Phi(x) + c) \quad (15)$$

Finally, the solution to the given ODE is

$$y = \frac{1}{e^{\frac{x^2}{2}} (-\Phi(x) + c)} \quad (16)$$

References

- [1] C. H. Edwards and D. E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th ed. Upper Saddle River, NJ: Pearson, 2000.